WASP-50b

R-1709

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-50_131102` · created 2026-08-02T15:14:25+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456598.718 +/- 0.091 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.05
Transit depth (Rp/Rs)^2	2.53 +/- 1.61 [%]
Orbital Inclination (inc)	89.6 +/- 2.89
Airmass coefficient 1 (a1)	0.93 +/- 0.63
Airmass coefficient 2 (a2)	0.1 +/- 0.37
Scatter in the residuals of the lightcurve fit is	73.3 %
Best Comparison Star	None
Optimal Aperture	2.64
Optimal Annulus	3.0
Transit Duration (day)	0.052 +/- 0.036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.159 ± 0.05 vs 0.1404 (deviation 0.36σ)
Duration fit vs published	0.052 d vs 0.0754 d (deviation 0.63σ)
Mid-time O–C	-5.9 min
Depth SNR	3.1
Residual scatter	73.3 %

Light curve

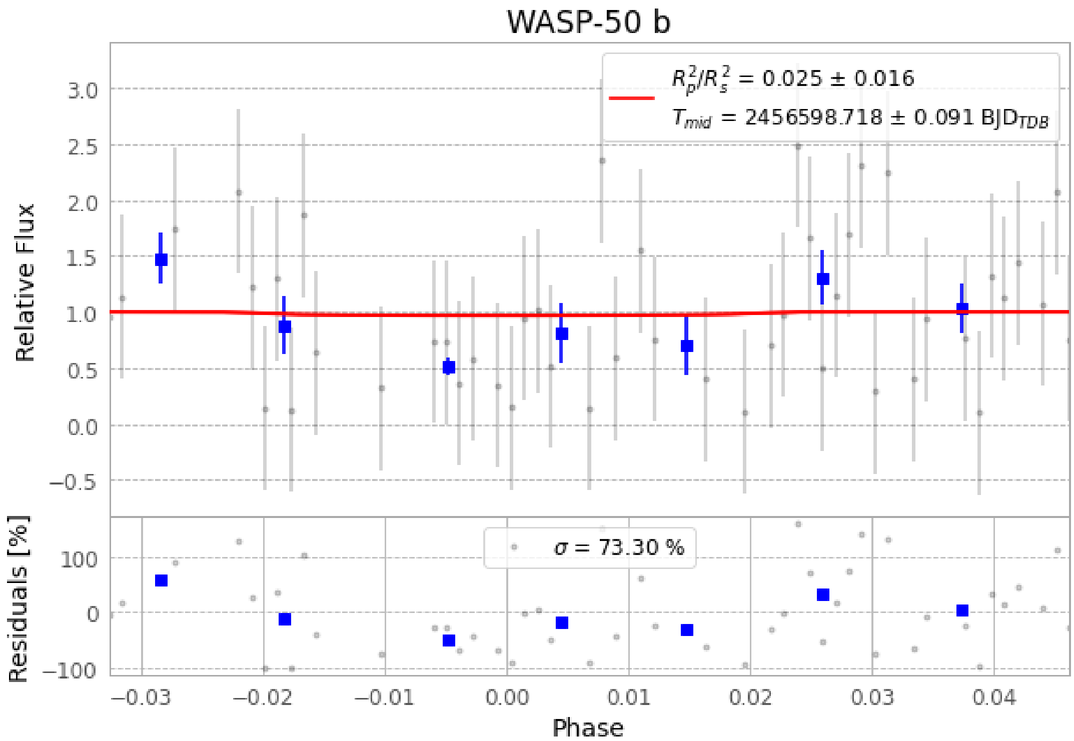

Figure 1 — FinalLightCurve_WASP-50 b_2013-11-01.png (SHA-256 9bd5240de50a2eab…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [128, 223], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 11c367ac5f12c20e…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

77 FITS frames (51 MB), `WASP-50131102032712.FITS` → `WASP-50131102071513.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-131102.log (0e33777db7b1…), FinalLightCurve_WASP-50 b_2013-11-01.png (9bd5240de50a…), FinalLightCurve_WASP-50 b_2013-11-01.csv (9ccf14204593…)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)